

DEPARTMENT OF MATHEMATICS & STATISTICS
MTH 211: THEORY OF STATISTICS (2025-26)
Midsemester Exam – Solutions

Question 1: Bivariate Uniform Rectangle

(a) Jointly Minimal Sufficient Statistic

Solution:

The joint pdf for a random sample $(X_1, Y_1), \dots, (X_n, Y_n)$ from the bivariate uniform distribution on $(0, \theta_1) \times (0, \theta_2)$ is:

$$f(\mathbf{x}, \mathbf{y} | \theta_1, \theta_2) = \prod_{i=1}^n \frac{1}{\theta_1 \theta_2} \mathbb{I}(0 < x_i < \theta_1) \mathbb{I}(0 < y_i < \theta_2)$$

$$f(\mathbf{x}, \mathbf{y} | \theta_1, \theta_2) = \frac{1}{\theta_1^n \theta_2^n} \prod_{i=1}^n \mathbb{I}(0 < x_i < \theta_1) \prod_{i=1}^n \mathbb{I}(0 < y_i < \theta_2)$$

Using the property of order statistics, we can rewrite the indicators:

$$f(\mathbf{x}, \mathbf{y} | \theta_1, \theta_2) = \frac{1}{\theta_1^n \theta_2^n} \mathbb{I}(x_{(n)} < \theta_1) \mathbb{I}(y_{(n)} < \theta_2) \mathbb{I}(x_{(1)} > 0) \mathbb{I}(y_{(1)} > 0)$$

By the **Factorization Theorem**, let:

$$g(x_{(n)}, y_{(n)}; \theta_1, \theta_2) = \frac{1}{\theta_1^n \theta_2^n} \mathbb{I}(x_{(n)} < \theta_1) \mathbb{I}(y_{(n)} < \theta_2)$$

$$h(\mathbf{x}, \mathbf{y}) = \mathbb{I}(x_{(1)} > 0) \mathbb{I}(y_{(1)} > 0)$$

Since g depends on the data only through $(X_{(n)}, Y_{(n)})$, $T = (X_{(n)}, Y_{(n)})$ is **sufficient**. To show it is **minimal**, consider the ratio of likelihoods at $\mathbf{x}$ and $\mathbf{u}$:

$$\frac{f(\mathbf{x}, \mathbf{y} | \theta_1, \theta_2)}{f(\mathbf{u}, \mathbf{v} | \theta_1, \theta_2)} = \frac{\mathbb{I}(x_{(n)} < \theta_1) \mathbb{I}(y_{(n)} < \theta_2)}{\mathbb{I}(u_{(n)} < \theta_1) \mathbb{I}(v_{(n)} < \theta_2)}$$

This ratio is independent of (θ_1, θ_2) if and only if $x_{(n)} = u_{(n)}$ and $y_{(n)} = v_{(n)}$, proving T is **jointly minimal sufficient**.

(b) Completeness for $\theta_1 = \theta_2 = \theta$

Solution:

Disprove: The statistic is Not Complete. We test if $E[h(T)] = 0$ for all $\theta > 0$ implies $h(T) = 0$ almost everywhere. Consider the function $h(T) = X_{(n)} - Y_{(n)}$. Since X_i and Y_i are independent and both follow $U(0, \theta)$, their maximums have the same PDF:

$$f_{X_{(n)}}(t) = f_{Y_{(n)}}(t) = \frac{nt^{n-1}}{\theta^n}, \quad 0 < t < \theta$$

. Calculating the expectations:

$$E[X_{(n)}] = \int_0^\theta t \left(\frac{nt^{n-1}}{\theta^n} \right) dt = \frac{n}{\theta^n} \left[\frac{t^{n+1}}{n+1} \right]_0^\theta = \frac{n}{n+1} \theta$$

$$E[Y_{(n)}] = \int_0^\theta t \left(\frac{nt^{n-1}}{\theta^n} \right) dt = \frac{n}{n+1} \theta$$

Therefore, the expectation of our chosen function is:

$$E[h(T)] = E[X_{(n)} - Y_{(n)}] = E[X_{(n)}] - E[Y_{(n)}] = 0 \quad \text{for all } \theta > 0$$

However, $h(T) = X_{(n)} - Y_{(n)}$ is clearly not zero with probability 1. Thus, $T = (X_{(n)}, Y_{(n)})$ is **not complete** for this restricted parameter space.

Common Mistakes :

1. Not noticing that only the n 'th order statistics depend on θ and writing $T = (X_{(1)}, X_{(n)}, Y_{(1)}, Y_{(n)})$.
2. Finding out the MSS for case (b) and checking that for completeness, which is not what the question asked. The question specifically asked to check the statistic found in part (a) for completeness.
3. Incorrectly calculating $f_{X_{(n)}}(x)$ and $E[X_{(n)}]$.

Question 2: Pokémon Populations

(a) Ditto (Power Function Distribution)

Solution:

i. Identify a Sufficient Statistic T: The joint density function for n independent positions is:

$$f(\mathbf{x}|\theta) = \prod_{i=1}^n \frac{2x_i}{\theta^2} \mathbb{I}(0 \leq x_i \leq \theta)$$

$$f(\mathbf{x}|\theta) = \frac{2^n \prod_{i=1}^n x_i}{\theta^{2n}} \mathbb{I}(\max(x_1, \dots, x_n) \leq \theta)$$

By the Factorisation Theorem, we decompose this into:

$$g(x_{(n)}, \theta) = \frac{1}{\theta^{2n}} \mathbb{I}(x_{(n)} \leq \theta) \quad \text{and} \quad h(\mathbf{x}) = 2^n \prod_{i=1}^n x_i$$

Thus, $T = X_{(n)} = \max(X_1, \dots, X_n)$ is the **sufficient statistic**.

ii. PDF of T and Proof of Completeness: The CDF of X is $F_X(x) = \int_0^x \frac{2u}{\theta^2} du = \frac{x^2}{\theta^2}$. The CDF of T is $F_T(t) = P(X_1 \leq t, \dots, X_n \leq t) = [F_X(t)]^n$:

$$F_T(t) = \left(\frac{t^2}{\theta^2} \right)^n = \frac{t^{2n}}{\theta^{2n}}, \quad 0 \leq t \leq \theta$$

The PDF of T is $f_T(t) = \frac{d}{dt} F_T(t)$:

$$f_T(t) = \frac{2nt^{2n-1}}{\theta^{2n}}, \quad 0 \leq t \leq \theta$$

To check **completeness**, let $E[g(T)] = 0$ for all $\theta > 0$:

$$\int_0^\theta g(t) \frac{2nt^{2n-1}}{\theta^{2n}} dt = 0 \implies \int_0^\theta g(t) t^{2n-1} dt = 0$$

Differentiating with respect to θ using the Leibniz Rule:

$$g(\theta)\theta^{2n-1} = 0 \implies g(\theta) = 0 \text{ for all } \theta > 0$$

Since $g(T) = 0$ almost everywhere, T is **complete**.

(b) Snorlax (UMVUE for μ^3)

Solution:

For $N(\mu, \sigma^2)$ where both parameters are unknown, the **complete sufficient statistics** are $(\bar{X}, S^2)$. We seek the UMVUE for $g(\mu) = \mu^3$. Start with the estimator $\bar{X}^3$ and find its expectation:

$$E[\bar{X}^3] = \mu^3 + 3\mu\text{Var}(\bar{X}) = \mu^3 + \frac{3\mu\sigma^2}{n}$$

To remove the bias term $\frac{3\mu\sigma^2}{n}$, note that $E[\bar{X}] = \mu$ and $E[S^2] = \sigma^2$. Since $\bar{X}$ and S^2 are independent for Normal samples:

$$E[\bar{X}S^2] = E[\bar{X}]E[S^2] = \mu\sigma^2$$

Thus, we can construct an unbiased estimator for μ^3 :

$$E\left[\bar{X}^3 - \frac{3\bar{X}S^2}{n}\right] = \left(\mu^3 + \frac{3\mu\sigma^2}{n}\right) - \frac{3\mu\sigma^2}{n} = \mu^3$$

By the **Lehmann-Scheffé Theorem**, since this estimator is unbiased and is a function of the complete sufficient statistics, it is the **UMVUE**:

$$\text{UMVUE} = \bar{X}^3 - \frac{3\bar{X}S^2}{n}$$

Common Mistakes :

1. Some of you have proved it to be not complete , So basics should be clear .
2. Though almost everyone correctly identified the complete statistics but many couldn't find the UMVUE

Question 3: Method of Moments (MOM)

Solution:

For a sample $X_1, \dots, X_n \sim \text{Gamma}(\alpha, \beta)$, we set estimate of the population moments equal to the sample moments:

$$\mu'_1 = E[X_1] = \alpha\beta, \quad \hat{\mu}'_1 = m'_1 = \bar{X}$$

$$\mu'_2 = E[X_1^2] = \text{Var}(X_1) + (E[X_1])^2 = \alpha\beta^2 + \alpha^2\beta^2, \quad \hat{\mu}'_2 = m'_2 = \frac{1}{n} \sum X_i^2$$

We can rewrite the second equation using the first:

$$\alpha\beta^2 + (\mu'_1)^2 = \mu'_2$$

$$\alpha\beta^2 = \mu'_2 - (\mu'_1)^2 = \mu_2$$

Now, we divide the variance equation by the mean equation:

$$\frac{\alpha\beta^2}{\alpha\beta} = \frac{\mu_2}{\mu_1}$$

so that,

$$\hat{\beta}_{MOM} = \frac{m_2' - (m_1')^2}{m_1'} = \frac{S_n^2}{\bar{X}}$$

Substituting $\hat{\beta}$ back into the mean equation:

$$\hat{\alpha}_{MOM} \left(\frac{S_n^2}{\bar{X}} \right) = \bar{X}$$

giving,

$$\hat{\alpha}_{MOM} = \frac{\bar{X}^2}{S_n^2},$$

where S_n^2 is the sample variance with divisor n .

Common Mistakes / Feedback :

1. Many of you are equating the population moments to the sample moments! This is wrong since you cannot equate fixed parameters (population moments) with functions of random variables (sample moments).
2. Though, ultimately may be you're writing the correct answer, some marks have been deducted for this conceptual mistake.

Question 4: Efficiency and CRLB

Solution:

i. Finding the CRLB for σ^2 :

Let $\theta = \sigma^2$. Since there is no DC offset, the mean $\mu = 0$ is known:

$$\log f(x|\theta) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \theta - \frac{x^2}{2\theta}$$

Differentiating with respect to θ :

$$\frac{\partial}{\partial \theta} \log f = -\frac{1}{2\theta} + \frac{x^2}{2\theta^2}$$

Taking the second derivative:

$$\frac{\partial^2}{\partial \theta^2} \log f = \frac{1}{2\theta^2} - \frac{x^2}{\theta^3}$$

Fisher Information for n independent readings is:

$$I(\theta) = -nE \left[\frac{\partial^2}{\partial \theta^2} \log f(X|\theta) \right] = -n \left(\frac{1}{2\theta^2} - \frac{E[X^2]}{\theta^3} \right)$$

Since $E[X^2] = \text{Var}(X) + (E[X])^2 = \theta + 0 = \theta$:

$$I(\theta) = -n \left(\frac{1}{2\theta^2} - \frac{\theta}{\theta^3} \right) = -n \left(\frac{1}{2\theta^2} - \frac{1}{\theta^2} \right) = \frac{n}{2\theta^2}$$

$$\text{CRLB} = \frac{1}{I(\theta)} = \frac{2\theta^2}{n} = \frac{2\sigma^4}{n}$$

ii. Efficiency of S^2 :

The sample variance is defined as $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$. For a normal distribution $N(0, \sigma^2)$, the variance of S^2 is:

$$\text{Var}(S^2) = \frac{2\sigma^4}{n-1}$$

Comparing the variance of the estimator to the CRLB:

$$\text{Var}(S^2) = \frac{2\sigma^4}{n-1} > \frac{2\sigma^4}{n} = \text{CRLB}$$

Since $\text{Var}(S^2) > \text{CRLB}$ for all $n > 1$, S^2 **does not attain** the CRLB.

Common Mistakes / Feedback :

1. It is good that a significant number were able to do this question correctly
2. But some weren't able to correctly find the fisher information leading to incorrect results.
3. We recommend to practice such questions

Question 5: Inverse Gaussian MLE

Solution:

The log-likelihood function is given by:

$$l(\mu, \lambda) = \frac{n}{2} \log \lambda - \sum_{i=1}^n \frac{\lambda(x_i - \mu)^2}{2\mu^2 x_i} + C$$

1. First Derivatives and Critical Points: Starting with μ :

$$\frac{\partial l}{\partial \mu} = -\frac{\lambda}{2} \sum_{i=1}^n \frac{\partial}{\partial \mu} \left[\frac{x_i}{\mu^2} - \frac{2}{\mu} + \frac{1}{x_i} \right]$$

$$\frac{\partial l}{\partial \mu} = -\frac{\lambda}{2} \sum_{i=1}^n \left[-\frac{2x_i}{\mu^3} + \frac{2}{\mu^2} \right]$$

$$\frac{\partial l}{\partial \mu} = \frac{\lambda}{\mu^3} \sum_{i=1}^n (x_i - \mu) = \frac{\lambda}{\mu^3} (n\bar{X} - n\mu)$$

Setting $\frac{\partial l}{\partial \mu} = 0 \implies \hat{\mu} = \bar{X}$.

Now for λ :

$$\frac{\partial l}{\partial \lambda} = \frac{n}{2\lambda} - \sum_{i=1}^n \frac{(x_i - \mu)^2}{2\mu^2 x_i}$$

Setting $\frac{\partial l}{\partial \lambda} = 0$ and substituting $\hat{\mu} = \bar{X}$:

$$\frac{n}{\lambda} = \sum_{i=1}^n \frac{x_i^2 - 2x_i\bar{X} + \bar{X}^2}{\bar{X}^2 x_i}$$

$$\begin{aligned}\frac{n}{\lambda} &= \sum_{i=1}^n \left(\frac{x_i}{\bar{X}^2} - \frac{2}{\bar{X}} + \frac{1}{x_i} \right) \\ \frac{n}{\lambda} &= \frac{n\bar{X}}{\bar{X}^2} - \frac{2n}{\bar{X}} + \sum_{i=1}^n \frac{1}{x_i} = \sum_{i=1}^n \frac{1}{x_i} - \frac{n}{\bar{X}} \\ \hat{\lambda} &= \frac{n}{\sum \frac{1}{x_i} - \frac{n}{\bar{X}}}\end{aligned}$$

2. Second Derivative Test (Verification of Maximum): We compute the Hessian matrix H at $(\hat{\mu}, \hat{\lambda})$. First, the second derivative with respect to μ :

$$\begin{aligned}\frac{\partial^2 l}{\partial \mu^2} &= \frac{\partial}{\partial \mu} \left[\frac{\lambda}{\mu^3} (n\bar{X} - n\mu) \right] \\ \frac{\partial^2 l}{\partial \mu^2} &= -\frac{3\lambda}{\mu^4} (n\bar{X} - n\mu) - \frac{n\lambda}{\mu^3}\end{aligned}$$

Evaluating at $\hat{\mu} = \bar{X}$:

$$\left. \frac{\partial^2 l}{\partial \mu^2} \right|_{\hat{\mu}} = 0 - \frac{n\lambda}{\bar{X}^3} < 0$$

Next, the second derivative with respect to λ :

$$\frac{\partial^2 l}{\partial \lambda^2} = \frac{\partial}{\partial \lambda} \left[\frac{n}{2\lambda} - \text{const} \right] = -\frac{n}{2\lambda^2} < 0$$

Finally, the mixed partial derivative:

$$\frac{\partial^2 l}{\partial \mu \partial \lambda} = \frac{\partial}{\partial \lambda} \left[\frac{\lambda}{\mu^3} (n\bar{X} - n\mu) \right] = \frac{1}{\mu^3} (n\bar{X} - n\mu)$$

Evaluating at $\hat{\mu} = \bar{X}$:

$$\left. \frac{\partial^2 l}{\partial \mu \partial \lambda} \right|_{\hat{\mu}} = 0$$

The Hessian matrix H is:

$$H = \begin{bmatrix} -\frac{n\lambda}{\bar{X}^3} & 0 \\ 0 & -\frac{n}{2\lambda^2} \end{bmatrix}$$

To prove $(\hat{\mu}, \hat{\lambda})$ is a local maximum, we show that the Hessian is ****negative definite****. For any non-zero vector $\mathbf{v} = (v_1, v_2)^T \in \mathbb{R}^2$:

$$\begin{aligned}\mathbf{v}^T H \mathbf{v} &= \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} -\frac{n\lambda}{\bar{X}^3} & 0 \\ 0 & -\frac{n}{2\lambda^2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \\ \mathbf{v}^T H \mathbf{v} &= -\frac{n\lambda}{\bar{X}^3} v_1^2 - \frac{n}{2\lambda^2} v_2^2\end{aligned}$$

Since $n, \lambda, \bar{X} > 0$, it follows that:

$$\mathbf{v}^T H \mathbf{v} < 0 \quad \forall \mathbf{v} \in \mathbb{R}^2, \mathbf{v} \neq \mathbf{0}$$

Because the quadratic form is strictly negative for all non-zero vectors, the Hessian is negative definite, confirming that $(\hat{\mu}, \hat{\lambda})$ is a **local maximum**.

Common Mistakes / Feedback :

1. Most of you could obtain the ML estimator of μ , but made mistakes in obtaining that of λ .
2. Just solving the likelihood equation/s is not obtaining the ML estimator/s. You need to check the second derivative condition, the negative definiteness of the Hessian Matrix.

Question 6: MLE for Exponential Distribution**Solution:****i. Obtaining the MLE of θ :**

The likelihood function for a sample $X_1, X_2, \dots, X_n$ is:

$$L(\theta) = \prod_{i=1}^n f(x_i|\theta) = \prod_{i=1}^n \theta e^{-\theta x_i} = \theta^n e^{-\theta \sum x_i}$$

Taking the natural logarithm to get the log-likelihood:

$$\ell(\theta) = \log L(\theta) = n \log \theta - \theta \sum_{i=1}^n x_i$$

Differentiating with respect to θ and setting to zero:

$$\frac{d}{d\theta} \ell(\theta) = \frac{n}{\theta} - \sum_{i=1}^n x_i = 0$$

Solving for θ gives the Maximum Likelihood Estimator (MLE):

$$\hat{\theta}_{MLE} = \frac{n}{\sum_{i=1}^n X_i} = \frac{1}{\bar{X}}$$

To verify it is a maximum, we check the second derivative:

$$\frac{d^2}{d\theta^2} \ell(\theta) = -\frac{n}{\theta^2} < 0$$

Thus, $\hat{\theta} = 1/\bar{X}$ is the MLE.

ii. Consistency of the MLE:

By the Weak Law of Large Numbers (WLLN), the sample mean $\bar{X}$ converges in probability to the population mean $E[X]$. For $X \sim \text{Exp}(\theta)$, $E[X] = 1/\theta$.

$$\bar{X} \xrightarrow{P} \frac{1}{\theta}$$

Since $g(u) = 1/u$ is a continuous function for $u > 0$, by the Continuous Mapping Theorem:

$$\hat{\theta} = \frac{1}{\bar{X}} \xrightarrow{P} \frac{1}{1/\theta} = \theta$$

Therefore, the MLE is a **consistent estimator** of θ .

iii. Asymptotic Distribution:

The Fisher Information for a single observation is:

$$I(\theta) = -E \left[\frac{\partial^2}{\partial \theta^2} \log f(X|\theta) \right] = -E \left[-\frac{1}{\theta^2} \right] = \frac{1}{\theta^2}$$

By the asymptotic properties of MLEs, $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{1}{I(\theta)}\right)$. Substituting $I(\theta) = 1/\theta^2$:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta^2)$$

Equivalently, for large n :

$$\hat{\theta} \sim AN\left(\theta, \frac{\theta^2}{n}\right)$$

Common Mistakes / Feedback :

1. Most of you could obtain the ML estimator. Remember, just solving the likelihood equation/s is not obtaining the ML estimator/s. You need to check the second derivative condition/s.
2. Many of you just wrote 'ML estimator is consistent', when you were supposed to prove it in this case! There are two very simple ways - either use the WLLN, or the sufficient conditions of consistency.
3. Some of you miscalculated $I(\theta) = \frac{n}{\theta^2}$ and hence got the asymptotic distribution wrong.

Question 7: Confidence Limit for θ

Solution:

i. Pivotal Quantity:

For the given sample from the distribution, let $Y_i = X_i/\theta$, $i = 1(1)n$. Then Y_i 's are iid $U(0, 1)$. We define the pivotal quantity $Y = X_{(n)}/\theta$. The distribution of Y is $Beta(n, 1)$.

ii. Finding the Confidence Limits:

Case(a): We seek two-sided confidence limits (k_1, k_2) such that $P(k_1 \leq Y \leq k_2) = 1 - \alpha$. Take $P(Y > k_2) = \frac{\alpha}{2}$ and $P(Y < k_1) = \frac{\alpha}{2}$. Then using the cdf of Y , we get $k_1 = (\frac{\alpha}{2})^{1/n}$ and $k_2 = (1 - \frac{\alpha}{2})^{1/n}$, so that $P\left((\frac{\alpha}{2})^{1/n} \leq X_{(n)}/\theta \leq (1 - \frac{\alpha}{2})^{1/n}\right) = 1 - \alpha$.

Finally, inverting the pivot, we get the $100(1 - \alpha)\%$ C.I for θ as $\left[\frac{X_{(n)}}{(1 - \frac{\alpha}{2})^{1/n}}, \frac{X_{(n)}}{(\frac{\alpha}{2})^{1/n}}\right]$

Case(b): For an alternative two-sided C.I., we take the natural upper bound of Y and consider $P(k \leq Y \leq 1) = 1 - \alpha$. Using the cdf of Y , we get $k = \alpha^{1/n}$.

Hence, $P\left(\alpha^{1/n} \leq X_{(n)}/\theta \leq 1\right) = 1 - \alpha$.

Inverting the pivot, we get the $100(1 - \alpha)\%$ C.I for θ as $\left[X_{(n)}, \frac{X_{(n)}}{\alpha^{1/n}}\right]$.

Common Mistakes / Feedback :

1. Many of you could obtain the natural upper bound (Case (b)) C.I.
2. Some of you tried with both unknown limits, but ended up solving them incorrectly. Note that correct limits can involve only $X_{(n)}$, α , n and any fixed constant, nothing else!
3. $-\infty$ as lower limit or ∞ as upper limit is also not accepted, since 2-sided CI has been asked.

4. Some of you missed the word 'two' - unfortunately!